

- 2.1 For an nMOS transistor, its I-V characteristics may be described by the first-order Shockley model,

$$I_{ds} = \begin{cases} 0 & V_{gs} < V_t \\ \beta(V_{GT} - \frac{V_{ds}}{2})V_{ds} & V_{ds} < V_{dsat} \\ \frac{\beta}{2}V_{GT}^2 & V_{ds} > V_{dsat} \end{cases},$$

for the cutoff, linear, and saturation regions of operation, respectively. β is given by

$$\beta = \mu_n C_{ox} \frac{W}{L},$$

where μ_n is electron mobility, C_{ox} is gate oxide capacitance per unit area, and W and L refer to the width and length of the transistor channel, respectively.

Plotting the I-V curve, using the piecewise function given above, for an nMOS transistor with a $\frac{W}{L}$ ratio of $\frac{4}{2}$, a gate oxide thickness of 100Å (used to calculate C_{ox}), a threshold voltage of 0.7V, and an electron mobility of $350 \frac{cm^2}{V \cdot s}$ for $V_{gs} = 0, 1, 2, 3, 4$, and 5V,

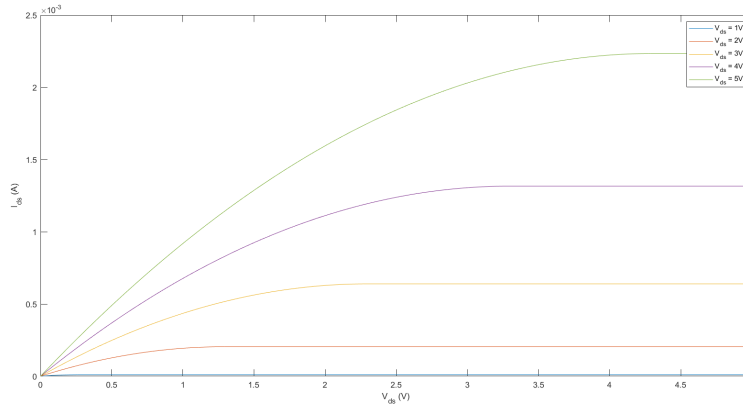


Figure 1: I-V curve for various values of V_{gs}

Note that at $V_{gs} = 0V$, the transistor is in cutoff mode, and no current flows between source and drain.

- 2.4 To find gate capacitance per micron of width $C_{permicron}$, we may use the formula for gate capacitance per unit area, $C_g = C_{ox}WL$. Dividing out the width of the gate leaves

$$\frac{C_g}{W} = C_{permicron} = C_{ox}L = 0.194 \frac{\mu F}{\mu m}.$$

2.9 The threshold voltage of any MOS transistor is defined as the minimum gate-to-source voltage V_{gs} that generates an inversion layer at the boundary between the insulating dielectric layer and the MOS transistor's body. In an nMOS transistor, the body is generally constructed with a p-type substrate (hence the "inversion" layer that forms as negatively charged particles are drawn to the body-dielectric boundary). The device's source and drain terminals, on the other hand, are constructed with an n-type material.

Thus, V_{gs} in an nMOS must be positive to generate a channel for current to flow between source and drain. The effect of raising the electrical potential of the body relative to ground would be to increase the threshold voltage, since a higher voltage relative to ground would now need to be provided to the gate to generate a conducting channel.

Conversely, lowering the electrical potential of the body relative to ground would have the opposite effect, decreasing the threshold voltage needed to generate a conducting channel.

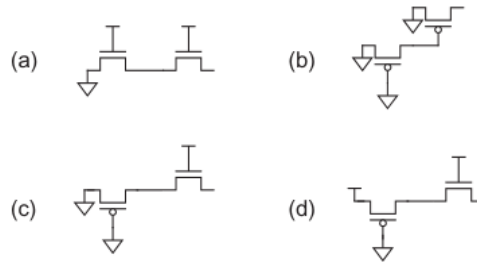


FIGURE 2.35 Pass transistor networks

Figure 2: Pass transistor networks

- 2.20
- The gate of both series-connected nMOS transistors are tied to V_{DD} , while the drain of the first is tied to ground. At the source of the second transistor, we may define the output voltage as $V_s = 0V$, since the gate-to-source voltage V_{gs} will remain well above V_{tn} when the source of both nMOS devices is pulled to ground.
 - Assuming both pMOS transistors share the same threshold voltage V_{tp} , the first transistor will provide a degraded voltage $|V_{tp}|$ to the second transistor's gate. Then, the second pass transistor will degrade the output by the same threshold drop, leaving a total output voltage of $V_s = 2|V_{tp}|$.
 - The initial pMOS pass transistor will pass a degraded voltage to the drain of the series-connected nMOS transistor, which will pass a "strong" voltage to its source terminal, resulting in a total output voltage of $V_d = V_s = |V_{tp}|$.

- d) In this configuration, the initial pMOS device passes a "strong" 1 to the drain of the nMOS device, which passes a degraded output voltage of $V_{DD} - V_{tn}$.
- 2.21 a) Given an input voltage $V_{in} = 0V$, the nMOS transistor, in its "on" state, will pass 0V to the source, resulting in a gate-to-source voltage $V_{gs} = 1.2V$. Since $V_{gs} > V_{tn}$, the output will not be degraded and $V_s = 0V$. This agrees with our notion that nMOS transistors pass a "strong" 0.
- b) Given $V_{in} = 0.6V$, V_{gs} will drop to 0.6V. Since V_{gs} is still greater than V_{tn} , the output will not be degraded, and $V_s = 0.6V$.
- c) At $V_{in} = 0.9$, V_{gs} drops to 0.3V, below the threshold voltage. Now, the output voltage is degraded, and $V_s = V_{DD} - V_{tn} = 0.8V$, a degraded 1.
- d) In this case, V_{in} is still greater than V_{tn} , and so the output remains at a degraded 1, with $V_s = 0.8V$.